# HONR 204-05H: Science & Society: Network Science

## Fall 2026 Syllabus

## Tue/Thu, 4:15 pm – 5:30 pm CT @ 324 Cuneo Hall

Last Updated: Oct 7

## Changelog

- New Project Demo date: Fri, Nov 6.
- Cal.com link for Prof. Sikdar

Please note that this syllabus may be subject to changes within the first few weeks of classes, mainly to enable minor corrections. Please check Sakai for any updates. Thank you for your understanding.

# Course Information at a Glance

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|---|---|
| **Course** | HONR 204-05H: Science & Society: Network Science |
| **Modality** | In-person @ 324 Cuneo Hall |
| **Duration** | Aug 24, 2026 – Dec 12, 2026 |
| **Lecture Time** | Tue/Thu, 4:15 pm – 5:30 pm CT |
| **Instructor** | Dr. Satyaki Sikdar |
| **Instructor Email** | ssikdar@luc.edu |
| **Instructor Office** | 211 Doyle Hall |
| **Office Hours** | Online or in-person by appointment via Cal |
| **Helpful Links** | Office Hours Gradescope Late Pass Form |

# Course Components

## Course Description

Networks are everywhere: friendships and social media connections, trade and financial systems, the Internet, and even the interactions inside living cells. Network Science provides a common language to study these complex, interconnected systems across disciplines. In this course, students will learn core concepts such as how to measure the structure of networks, identify important nodes and communities, and model processes like the spread of information or contagion. The class blends theory with real-world examples from computer science, sociology, political science, and biology, giving students practical tools to analyze and understand the connected world around them. By the end of the course, students will see how network science helps explain phenomena ranging from viral marketing campaigns to political polarization to the resilience of infrastructure systems.

## Learning Outcomes

Upon successful completion of this course, you will be able to:

- Define and Classify Network Structures: Identify and describe the fundamental components of a network (nodes and edges) and distinguish between different types of networks across social, biological, and technological domains.
- Quantify Network Topology: Calculate and interpret key structural metrics, including degree distribution, clustering coefficients, and path lengths, to characterize the "shape" of a complex system.
- Identify Influence and Community: Apply mathematical measures of centrality (e.g., betweenness, closeness) to identify influential nodes and utilize algorithms to detect communities or clusters within large datasets.
- Model Diffusion and Contagion: Simulate and analyze how information, behaviors, or diseases spread through a population using established models of network dynamics.

- Utilize Computational Tools: Employ practical software tools and "common language" frameworks to visualize, manipulate, and extract insights from real-world network data.
- Evaluate System Resilience: Analyze how network architecture contributes to the robustness of infrastructure, financial systems, and biological organisms against random failures or targeted attacks.

## Deliverables

### *Problem Sets*

Problem sets will be released on **Fridays at 1 pm** and due the following **Friday at 11:55 pm**. The *lowest-scoring* problem set will be dropped when calculating the final grade. Over the semester, 4-5 problem sets will be released.

### *Quizzes*

There will be **4** timed in-person quizzes administered during class hours. There may be some overlap in content across quizzes. Each quiz will be worth 20-30 points and contain a mix of multiple-choice and short-answer questions. They will be closed book and closed internet, but you can use your class notes. You are expected to solve the problems on your own. Collaborating with other students or using other resources will be considered an Honor Code violation.

### *Final Exam*

There will not be a final exam for this class. The final project presentations will instead be scheduled based on the university's final exam schedule.

### *Participation*

After each lecture, you will be asked to write a brief reflection essay (minimum 3 sentences) covering the topics discussed. These will be administered on **Sakai** and will be due **24 hours** after the end of the lecture. These will be graded primarily for completion, and count towards your Participation grade. The **three lowest-scoring assignments** will be dropped while calculating the final grade.

### *Course Project*

There will be a course project in the second half of the course. You will work in teams of 2-3 with your classmates. There will be checkpoints to track each group's progress. A final project report and an 8-10 min presentation are due during the week of final exams.

## Tools

### *Sakai*

Sakai is the online course management system we will use to run the class. All course content, including the syllabus, lecture videos, assignments, solutions, and other course materials, will be posted here. It will also host your gradebook, which will be updated at least once a month. Please check the Sakai page at least once a week.

*Gradescope*

We will use Gradescope to administer the weekly problem sets. You can access Gradescope via the Assignments page on Sakai. For more information about Gradescope, please visit this website.

*Microsoft Teams*

The preferred way to contact the course staff via DMs on Teams.

## Tentative Schedule

This schedule may be altered to suit the needs of the class. Please check the Sakai page for the latest updates.

| Week # | Week of | Topic | Deliverables |
|---|---|---|---|
| 1 | Aug 24 | Course Introduction, Networks 101 | |
| 2 | Aug 31 | Small World Effect, Strong & Weak Ties | PS 1 |
| 3 | Sep 7 | Homophily, Affiliation Networks | |
| 4 | Sep 14 | Friendship Paradox, Signed Networks. **Quiz 1.** | |
| 5 | Sep 21 | Signed Networks | |
| 6 | Sep 28 | Centrality Measures | PS 2. Project Proposals. |
| 7 | Oct 5 | No class Tue, Oct 6. Web & Power Laws. | |
| 8 | Oct 12 | Preferential Attachment. **Quiz 2.** | |
| 9 | Oct 19 | Clusters | PS 3 |
| 10 | Oct 26 | Cascades | |
| 11 | Nov 2 | Cascades. **Quiz 3** | Project Demo |
| 12 | Nov 9 | Epidemics | |
| 13 | Nov 16 | Epidemics | PS 4 |
| 14 | Nov 23 | Epidemics II. No Class on Thu, Nov 26. (Thanksgiving) | |
| 15 | Nov 30 | **Quiz 4**. Final Project Presentations. | |
| 16 | Dec 7 | Project Presentations during the Final Exam slot. | Final Project Report |

# Requirements and Expectations

## Course Textbooks

**Networks, Crowds, and Markets: Reasoning About a Highly Connected World** (required)
David Easley and John Kleinberg
ISBN: 978-0-521-19533-1
The book is freely available online and can be accessed using this link.

**Network Science** (optional)
Albert-László Barabási
The book is *freely* available online and can be accessed using this link.

Additional references may be used during the course which would be listed on Sakai.

## What I expect of you

- Be fully present in class, abide by the Loyola University Honor Code, be kind to yourself, fellow students, and staff, and inform me promptly if you can't meet course obligations.
- Communicate professionally in all class-related communications.
- Complete assigned activities thoughtfully and on time.
- Enjoy learning about Network Science.

## What you can expect of me

- I provide well-organized and reasonable coursework to help you schedule your study time.
- I provide regular graded activities to track your progress.
- I offer advanced guidance on grading expectations.
- I grade work promptly and fairly.
- I provide necessary support and training for classroom success.

# Evaluation and Grading

## Grade Distribution

| Component | Weight |
|---|---|
| Problem Sets | 25% |
| Quizzes | 40% |
| Course Project | 25% |
| Participation | 10% |
| *Total* | *100%* |

## Grading Scale

| A | A– | B+ | B | B– | C+ | C | C– | D+ | D | F |
|---|---|---|---|---|---|---|---|---|---|---|
| 94+ | 90+ | 87+ | 84+ | 80+ | 77+ | 74+ | 70+ | 67+ | 60+ | <60 |

## Grade Change and Regrade Requests

Any requests for a grade change on an assignment or a quiz must be made within **seven** days when the grades are published. Any clerical errors while grading will be happily corrected. All change requests for problem sets must be made on **Gradescope**.

# Course Policies

## General

- Please be kind to yourself, your fellow students, and the course staff.
- A Jesuit education emphasizes learning to respect the rights and opinions of others. To do this, please respect others by (1) allowing all classmates to express their opinions without fear of ridicule and (2) refraining from using profanity or making objectionable (gendered, racial, or ethnic) comments, especially those directed at a classmate.
- Each instructor receives a class roster with the student's legal name. I'm happy to honor your request to address you by an alternate name and/or gender pronoun.
- Life can be unpredictable, and unexpected problems may arise that affect your life and work. If you need to address any issues related to this class, please contact Prof. Sikdar. If you have a problem with Prof. Sikdar, please contact the Computer Science department chair, Prof. George Thiruvathukal.
- I'm committed to the principle of universal learning, which means that our classroom, virtual spaces, practices, and interactions are as inclusive as possible. Mutual respect, civility, and the ability to listen and observe others are essential for universal learning. Active, thoughtful, and respectful participation in all aspects of the course will make our time together productive and engaging.

## Syllabus Changes

This syllabus communicates the intended structure and policies of the course. Changes may become necessary as the semester progresses. Any material change will be announced in advance in class and posted on Sakai. University policies in effect during the semester take precedence where applicable.

## Attendance

The class will be conducted in-person. You are expected to attend all lectures and actively participate in class. At Loyola, faculty members will make efforts to accommodate students if the observance of a major religious holiday interferes with a student's academic work. If you are unable to attend a class, participate in a test or quiz, be present for a presentation, or complete an assignment on a specific day because of the observance of a religious holiday, you will be excused and provided the opportunity to make up the work. You will continue to be responsible for all assigned work and should notify the professor in advance through Loyola email of the religious observance(s) that conflict with your classes.

## Late Work and Make Up

Life often gets in the way of study. To accommodate that, you have 2 Late Passes with no questions asked. No late work will be accepted without late passes.

To use a Late Pass, you must:

- Notify Prof. Sikdar **before** the assignment is due. Any time before the deadline is fine. The notification must be made via the late pass submission form. Your request will be approved as long as you have late passes remaining. Thus, you do not need to wait for a response.
- Turn in the assignment within **3 calendar days** of the original due date. Please include a note that it is being submitted late with a Late Pass.
- You may only use one Late Pass per assignment.
- The Late Passes **will not** be applied automatically or retroactively to a past assignment.

## Academic Integrity

Academic integrity is the pursuit of scholarly activity in an open, honest, and responsible manner. Academic integrity is a guiding principle for all academic activity at Loyola University Chicago, and we expect all members of the University community to act under this principle. Failing to meet the following academic integrity standards seriously violates personal honesty and the academic ideals that bind the University into a learning community. These standards apply to both individual and group assignments.

Individuals working in a group may be held responsible if one of the group members has violated one or more of these standards.

1. Students may not plagiarize.
2. Students may not fabricate data.
3. Students may not collude or cheat.
4. Students may not facilitate academic misconduct.

Read through Loyola's full statement on Academic Integrity here. These policies fully apply to this course. The penalty for task-wise academic misconduct is losing all the task's points. Multiple events of misconduct will be reported through the proper channels.

## Using Generative AI tools

Academic integrity is a guiding principle for academia. You may not submit others' work without attribution, fabricate information, or facilitate misconduct of others. In this course, students may use AI tools for activities such as brainstorming and proofreading but must **never** submit AI-generated content as their own work, without attribution. **Any use of AI must be disclosed and properly cited** to maintain academic integrity and support the development of individual writing skills, otherwise sanctions will result. Remember, use it but cite it!

## Recording Notification

This course lectures may be recorded and distributed to you in a secure environment. You can watch these recordings on your computer, tablet, or smartphone. In Sakai, find the lecture video links on the course schedule. Your questions and comments may be recorded. Except for faculty and staff who require access, no content will be shared with others without your permission. Loyola University Chicago and your instructor jointly copyright these recordings. Posting them to other websites or without permission may result in disciplinary action and civil prosecution.

## Privacy Statement

Recording online and face-to-face activities promotes open conversations and addresses concerns about sharing class comments. Faculty and students can use recordings solely for internal class purposes during the course. Recordings will be mentioned in the course syllabus. Instructors can use recordings with consent or remove student activity. They may retain recordings for personal use.

## Intellectual Property

All course materials, including lectures, notes, and PowerPoints, are the professor's intellectual property. They may not be shared without written permission. Recordings of lectures require consent, which can only be used for review and not distributed. Similarly, I won't share or distribute your work without your permission.

## Statement of Intent

By remaining in this course, students agree to accept this syllabus as a contract and to abide by the guidelines outlined in the document. Students will be consulted should there be a necessary change to the syllabus.

## Notice of Reporting Obligations for Responsible Campus Partners

As an instructor, I am a Responsible Campus Partner (RCP) under Loyola's Comprehensive Policy and Procedures for Addressing Discrimination, Sexual Misconduct, and Retaliation. While my goal is for you to be able to engage fully and authentically with our course material through class discussions and written work, I also want to be transparent that as an RCP I am required to report certain disclosures of sexual misconduct (including sexual assault, sexual harassment, intimate partner and/or domestic violence, and/or stalking) to the Office for Equity & Compliance (OEC). As the University's Title IX office, the OEC coordinates the University's response to reports and complaints of sexual misconduct (as well as discrimination of any kind) to ensure students' rights are protected.

As an instructor, I also have an obligation under Illinois law to report disclosures of or suspected instances of child abuse or neglect.The University maintains such reporting requirements to ensure that any student who experiences sexual/gender-based violence receives accurate information about available resources and support. Such reports will **not** generate a report to law enforcement

(no student will ever be forced to file a report with the police). Additionally, the University's resources and support are available to all students even if a student chooses that they do not want any other action taken. If you have any questions about this policy, you are encouraged to contact the OEC at equity@luc.edu or 773-508-7766.
If you ever wish to speak with a confidential resource regarding gender-based violence, I encourage you to call The Line at 773-494-3810. The Line is staffed by confidential advocates from 8:30am-5pm M-F and 24 hours on the weekend when school is in session. Advocates can provide support, talk through your options (medical, legal, LUC reporting, safety planning, etc.), and connect you with resources as needed – *without* generating a report or record with the OEC.

## Student Accommodations

Loyola University provides reasonable accommodations for students with disabilities. Any student requesting accommodations related to a disability or other condition is required to register with Student Accessibility Center (SAC), located in Sullivan Center, Suite 117. Professors receive the accommodation notification from SAC via Accommodate. Students are encouraged to meet with their professor individually in order to discuss their accommodations. All information will remain confidential. Please note that in this class, software may be used to record class lectures in order to provide equal access to students with disabilities. Students approved for this accommodation use recordings for their personal study only and recordings may not be shared with other people or used in any way against the faculty member, other lecturers, or students whose classroom comments are recorded as part of the class activity. Recordings are deleted at the end of the semester. For more information about registering with SAC or questions about accommodations, please contact SAC at 773-508-3700 or SAC@luc.edu.

## Mental Health

Each year, 5-20% of US residents contract the flu, while around 7% experience a major depressive episode. Depression is as common as the flu, yet one is socially stigmatized while the other isn't. As your professor, if mental health issues interfere with your ability to function, you can ask for short-term extensions or accommodations, just as you might for a physical illness. If you're struggling, please reach out. I'll work with you to minimize the impact on your performance.
If health problems, life stressors, or emotional difficulties interfere with your academic or personal success, contact the Wellness Center. They offer free, confidential healthcare services, crisis intervention, individual counseling, and group. You can make an appointment online or call 773-508-2530 for appointments or 773-508-8883 to speak with a nurse. If your medical or mental health condition requires ongoing academic accommodation, please register with the Student Accessibility Services.